

KRATWISH SAGDEO

Software Engineer

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Summary

Software Engineering undergraduate specializing in Cyber Security with hands-on experience in building scalable applications, secure systems, and real-time architectures. Proficient in Java, Python, and Flutter with strong foundations in data structures, algorithms, and system design. Experienced in cloud-integrated development, machine learning systems, and secure software practices, with a focus on building resilient, high-performance solutions aligned with modern software engineering standards.

Skills

Programming: Java, Python, Dart

Frameworks & Technologies: Flutter, FastAPI, Firebase, REST APIs

Cloud & Databases: AWS, Cloud Firestore, MongoDB, SQL

Software Engineering: Data Structures & Algorithms, OOP, System Design, API Design, Debugging

Security & Tools: Snort (IDS/IPS), OWASP ZAP, Nikto, Bandit, UFW, Postman, Git

Concepts: CI/CD, Agile Development, Application Security, Machine Learning, Data Processing

Experience

Full Stack App Developer Intern

Feb 2026 – Present

CodeRoofs – IT Solution

- Architecting and developing scalable mobile applications using Flutter with modular and reusable component design.
- Designed real-time features including chat systems and structured Firestore data models supporting concurrent users.
- Implemented secure authentication and role-based access control improving application security and data integrity.
- Integrated REST APIs with proper validation, error handling, and optimized state management across layers.
- Improved application performance and reduced response latency through efficient database queries and optimized workflows.

Mobile Application Developer Intern

Aug 2025 – Nov 2025

Sanjivani Gramin and Shahari Vikas Prasikshana Samiti

- Developed and maintained cross-platform Flutter modules aligned with production requirements.
- Integrated third-party APIs and Firebase services to enable real-time updates and notifications.
- Collaborated in debugging and improving UI performance across multiple application screens.
- Enhanced application stability through systematic testing, issue tracking, and resolution workflows.

Projects

AI-Powered Interview Platform | *Next.js, FastAPI, LLM, AWS*

2026

- Built a production-ready AI interview system simulating real-world interviews using LLMs and cloud-based architecture.
- Designed multi-stage evaluation pipeline including speech analysis, semantic evaluation, and adaptive questioning.
- Implemented RAG-based architecture with embeddings to generate context-aware questions and feedback.
- Reduced response latency to < 2 **seconds** through optimized API workflows and efficient backend design.
- Improved evaluation accuracy by **30–40%** over keyword-based systems using semantic similarity techniques.
- Enabled adaptive interview flow increasing user engagement and personalization by **40%**.

Hybrid ML-Based Intrusion Detection System | *Python, FastAPI*

2025

- Developed an IDS pipeline processing **30k+ network flows** with real-time detection capabilities.
- Engineered 40+ features and trained hybrid ML models achieving **94% accuracy** and **0.92 F1-score**.
- Reduced false positive rate to < 5 using ensemble learning techniques.
- Integrated signature-based and ML-based detection improving threat detection rate by **18%**.
- Deployed system using FastAPI enabling scalable, low-latency inference for network monitoring.

Education

B.Tech in Computer Science Engineering (Cyber Security)

June 2023 – Present

Ramdeobaba College of Engineering and Management

CGPA: 9.06 / 10